

Savitribai Phule Pune University



Department of Technology Lab Manual Time Series Analysis

Course Title: Time Series Analysis & Forecasting

Target Audience: UG/PG Students (Data Science / Analytics)

Tools: Python, Pandas, NumPy, Statsmodels, Scikit-learn, Matplotlib, Seaborn

Link: [Link](#)

LAB 1: Data Understanding, Cleaning & Preparation

Objective:

- To load time series data and perform data cleaning operations required for forecasting.
- To handle missing values, outliers, incorrect formatting, and ensure correct time index setup.

Expected Learning Outcomes:

- After completing this lab, students will be able to:
- Import a time series dataset and clean it for analysis
- Convert date column to a proper DateTime index
- Identify and fix missing values / duplicates / outliers
- Understand the importance of data preparation before modeling

Questions / Tasks:

Q1. Load the attached time series dataset into Python.

- a) Display the first and last 10 rows.
- b) Convert the date column to DateTime type and set it as index.

Q2. Perform data cleaning on the dataset.

- a) Check and handle missing values (explain the method chosen).
- b) Detect and treat outliers (justify the approach).
- c) Check for duplicates and remove if found.

Q3. Plot the cleaned time series and write any 2 observations.

Viva Questions.

LAB 1: Data Cleaning & Preparation

1. What is a time series? Give two real-life examples.
 2. Why do we convert the date column to DateTime type?
 3. What happens if we don't set the date as an index in a time series dataset?
 4. How do missing values affect forecasting models?
 5. What methods can be used to handle missing values in time series data?
 6. How can we detect and treat outliers in time series data?
 7. Why is it important to clean data before applying forecasting models?
 8. What is the difference between "data cleaning" and "data preprocessing"?
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LAB 2: Time Series Decomposition & Exploratory Data Analysis (EDA)

Objective:

- To decompose the time series into its components: Trend, Seasonality, and Residual.
- To interpret each component for better understanding of data patterns.

Expected Learning Outcomes:

- Understand how decomposition helps in analyzing time series
- Differentiate between trend, seasonal, and irregular components
- Perform both additive and multiplicative decomposition

Questions / Tasks:

Q1. Perform **Additive Decomposition** on the dataset.

- a) Plot the components (Trend, Seasonality, Residual).
- b) Write 2 interpretations from the decomposed plots.

Q2. Perform **Multiplicative Decomposition** and plot the components.

- a) Compare Additive vs Multiplicative and mention which fits your dataset better and why.

Q3. Explain the difference between Trend, Seasonality, Cycle, and Irregular components with examples.

Viva Questions.

LAB 2: Time Series Decomposition

1. What are the main components of a time series?
 2. Define Trend, Seasonality, and Noise.
 3. What is the difference between Additive and Multiplicative decomposition?
 4. When do we prefer additive decomposition?
 5. What do you observe in the seasonal component of your dataset?
 6. What does the residual component represent?
 7. Why is decomposition important before building forecasting models?
 8. Can decomposition help in detecting anomalies? How?
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LAB 3: Stationarity Analysis using ADF & KPSS Tests

Objective:

- To check stationarity of the time series using statistical tests (ADF & KPSS).
- To analyze results and determine whether the data is stationary or not.

Expected Learning Outcomes:

- Perform ADF and KPSS tests
- Interpret p-values and test statistics
- Decide whether differencing or transformation is needed

Questions / Tasks:

Q1. Plot rolling mean and rolling standard deviation for the series.

a) Comment on whether the plot indicates stationarity.

Q2. Perform **ADF Test** and report:

- ADF Statistic
- p-value
- Inference (Stationary or Not?)

Q3. Perform **KPSS Test** and report:

- KPSS Statistic
- p-value
- Inference (Stationary or Not?)

Q4. Based on both results, conclude whether your series is stationary or non-stationary. Why do we need to check both ADF & KPSS?

Viva Questions.

LAB 3: Checking Stationarity (ADF & KPSS Tests)

1. What is stationarity?
 2. Why is stationarity important in time series modeling?
 3. What does the Augmented Dickey-Fuller (ADF) test check for?
 4. What does the KPSS test check for?
 5. How do you interpret the p-value in ADF and KPSS tests?
 6. Why do we use both ADF and KPSS tests instead of just one?
 7. If ADF says the series is stationary but KPSS says it's not, what does that mean?
 8. What visual methods can help identify non-stationarity?
 9. What are the two main types of non-stationarity in time series data?
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LAB 4: Stationarization Techniques – Differencing, Transformation & Box-Cox

Objective:

- To convert a non-stationary time series into a stationary one.
- To apply differencing, transformations, and Box-Cox to stabilize variance and mean.

Expected Learning Outcomes:

- Apply 1st order and seasonal differencing
- Use log/Box-Cox transformations to stabilize data
- Confirm stationarity after transformations

Questions / Tasks:

Q1. Apply **First Order Differencing** to the series.

a) Plot the result and state if stationarity is improved.

Q2. Apply **Seasonal Differencing (if seasonal)** and plot the result.

Q3. Apply Log Transformation and Box-Cox Transformation.

- a) Plot before and after transformation.
- b) Which transformation was more effective and why?

Q4. Re-run ADF & KPSS on the transformed data.

- a) Present results in a comparison table (Before vs After).
- b) State which transformation achieved stationarity.

Viva Questions.

LAB 4: Making the Series Stationary (Differencing, Transformation & Box-Cox)

1. What is differencing?
 2. What is the purpose of applying differencing to a time series?
 3. What does the “order of differencing” represent in ARIMA?
 4. What is the purpose of log transformation in time series?
 5. How does the Box-Cox transformation differ from log transformation?
 6. What kind of data is suitable for Box-Cox transformation?
 7. What happens if you over-difference in a time series?
 8. After transformation, how do you confirm that the data has become stationary?
 9. Why do we sometimes need both differencing and transformation together?
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LAB 5: Time Series Forecasting using AR, MA, ARMA & ARIMA Models

Objective:

- To build classical forecasting models and compare forecasting accuracy.
- To understand the impact of p, d, q parameters on model performance.

Expected Learning Outcomes:

- Identify p & q using ACF and PACF plots
- Build AR, MA, ARMA, and ARIMA models
- Forecast future values and evaluate model performance

Questions / Tasks:

Q1. Plot ACF and PACF and determine values of p and q.

Q2. Build the following models and forecast the next 12 periods:

- a) AR Model
- b) MA Model
- c) ARMA Model
- d) ARIMA Model

Q3. For each model, report:

- Model order parameters
- RMSE or MAPE
- Forecast plot

Q4. Which model performed best? Justify using performance metrics and visual accuracy.

Viva Questions.

LAB 5: AR, MA, ARMA & ARIMA Models

1. What do p, d, and q represent in ARIMA?
 2. What is an AR (Auto-Regressive) model?
 3. What is an MA (Moving Average) model?
 4. How does ARMA differ from ARIMA?
 5. Why do we include differencing (d) in ARIMA but not in ARMA?
 6. What are ACF and PACF plots used for?
 7. How do we decide values for p and q using ACF and PACF plots?
 8. What metric would you use to evaluate an ARIMA model?
 9. What is residual analysis and why is it important after fitting a model?
 10. What is the main assumption about residuals in a good time series model?
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LAB 6: Time Series Forecasting using Machine Learning Models

Objective:

- To apply ML regression models for future prediction of time-dependent data.
- To evaluate and compare ML vs statistical time series models.

Expected Learning Outcomes:

- Perform feature engineering for time series (lags, rolling stats)
- Train ML models such as Linear Regression, Random Forest, or XGBoost
- Compare results against classical ARIMA-based models

Questions / Tasks:

Q1. Create the following features:

- a) lag features (Lag1, Lag2, Lag3)
- b) rolling mean/rolling std (any 1)

Q2. Train any **two ML models** (e.g., Linear Regression, Random Forest, XGBoost).

- a) Predict next values
- b) Plot Actual vs Predicted

Q3. Evaluate and compare models using RMSE or MAPE.

- a) Which model gave better accuracy?
- b) Mention one strength and one limitation of ML forecasting.

Viva Questions.

LAB 6: Forecasting using Machine Learning

1. How is machine learning different from classical time series models?
 2. Why do we avoid random shuffling in time series data?
 3. What is a lag feature? Give an example.
 4. Why do we create rolling mean or rolling std features?
 5. Can Linear Regression be used for time series forecasting? How?
 6. What is the disadvantage of using Random Forest for time series forecasting?
 7. What is overfitting in the context of forecasting models?
 8. Which performance metrics are commonly used for time series predictions?
 9. Why might an ML model outperform ARIMA in some cases?
 10. Why might an ARIMA model outperform ML in others?
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Viva Questions

1. Why do we perform stationarity checks before applying ARIMA?
2. What key difference exists in assumptions between ARIMA and ML models?
3. When would you prefer Box-Cox transformation over log transformation?
4. Why can't we shuffle data while training ML models for time series?

Some More

1. Explain the complete workflow of a time series forecasting project.
2. What are the differences between ARIMA and SARIMA models?
3. What is seasonality? How can you detect it in a dataset?
4. What does differencing do mathematically to a time series?
5. What happens if the data has trend and seasonality together?
6. What are the limitations of ARIMA models?
7. How does the Box-Cox transformation help stabilize variance?
8. How can you identify if your model is underfitting or overfitting?
9. What are residuals, and what do they tell you about your model?
10. Compare ARIMA, SARIMA, and LSTM models for forecasting.
11. What's the main advantage of using machine learning over statistical models?
12. How would you handle missing time steps (e.g., missing months) in time series?
13. What role does feature engineering play in ML-based forecasting?
14. How can you visualize model accuracy in forecasting problems?
15. How do you decide how far ahead to forecast?